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Effective Complaint Detection in Financial Services through Complaint, Severity, Emotion and Sentiment Analysis

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Abstract

Sentiment analysis plays a crucial role in understanding customer feedback and opinions within the financial domain. This research addresses the problem of sentiment-aware complaint identification using the FINancial Complaint CORpus (FINCORP) dataset, which provides annotated complaints along with sentiment, emotion and severity labels. The dataset encompasses various sentiment classes including Negative, Neutral, and Positive, enabling a wide understanding of customer sentiment towards financial products and services. Leveraging natural language processing (NLP) techniques such as text classification and sentiment analysis, this study aims to develop models capable of accurately detecting and categorizing sentiment, emotion of customers and the severity level of financial complaints. Effectively analyzing these factors allows financial institutions to gain valuable insights into customer satisfaction levels, pinpointing areas for improvement and enhance both customer experience and service quality. This research contributes to the advancement of sentiment analysis methodologies in the financial domain, facilitating better decision-making and customer relationship management strategies. DistilBERT, Glove and RoBERTa has outperformed in classifying the sentiment,emotion and severity of complaints,achieving F1-scores of 0.71, 0.74, and 0.86, respectively using logistic regression and Bi-LSTM classifiers.

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Keywords: Natural Language Processing; Sentiment Analysis; Financial complaint detection; Word Embedding; Machine Learning; Deep Learning

1. Introduction

In the digital era, financial complaints have become significantly prevalent, driving individuals to express their grievances on social media platforms. From banking mistakes to investment discrepancies, consumers utilize platforms like Twitter to bring out their dissatisfaction and seek resolution. The volume and variety of these financial complaints underscore the urgent need for effective classification methods to assist companies in promptly and efficiently addressing issues. In this context, leveraging advanced techniques such as natural language processing

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(NLP) [1], machine learning (ML) [2] and deep learning (DL) [3] becomes imperative to accurately categorize and analyse the magnitude of complaints flooding on online channels. By harnessing these technologies, companies can not only enhance their complaint management processes but also gain valuable insights into consumer sentiment[4] and behaviour in the financial domain.

Financial complaints wield significant influence, both on individual consumers and the broader financial landscape. These grievances can range from billing discrepancies and unauthorized charges to issues with loan approvals and investment advice. At the micro level, unresolved complaints can erode customer trust, leading to churn and tarnished reputations for financial institutions. Moreover, they can escalate into legal disputes, resulting in financial penalties and regulatory scrutiny. On a macro scale, aggregated complaints shed light on systemic issues within the industry, prompting regulatory interventions and policy reforms. Beyond the immediate financial implications, unresolved complaints can also have intangible consequences, such as negative publicity and damage to brand reputation. Thus, understanding, analysing and effectively addressing financial complaints are paramount for fostering trust, ensuring compliance and sustaining a healthy financial ecosystem.

This study employs a comprehensive methodology, merging machine learning, deep learning,ensemble methods [5] using various embedding to classify financial complaints within the FINCORP data for a multi-class task[6]. Our primary aim is to accurately categorize the severity level, emotion[7] and sentiment conveyed in each complaint tweet [8]. Through the utilization of these advanced methodologies, we strive to refine the precision of complaint classification, thereby making significant contributions to the advancement of more efficient complaint management systems. Additionally, our research seeks to deepen insights into consumer behaviour and sentiment within the financial domain, providing valuable knowledge for financial institutions to enhance customer satisfaction and service quality. By harnessing these techniques, we expect significant progress in comprehending and addressing the diverse view of financial grievances expressed across social media platforms such as Twitter.

Building upon the foundations of word embedding techniques like FastText which was introduced as an extension of Word2Vec. This method represents each word as a bag of character n-grams, enabling it to handle out-of-vocabulary words and capture morphological information more effectively. In recent years, the development of Bidirectional Encoder Representations from Transformers (BERT) has revolutionized the field of NLP including sentiment classification. BERT is a transformer-based neural network that learns context-aware word vectors, meaning that the same word can have different vector representations depending on the context in which it appears. This has led to state-of-the-art results in many NLP tasks as BERT that has the ability to capture the nuances and complexities of language more accurately than previous word embedding techniques.

The highlights in this paper is summarized as below:

- Integrating diverse ML and DL models for multi-task financial complaint identification encomposing severity, sentiment and emotion.
- Study on the impact of various embedding techniques on the built-in model.
- Addresses the challenge of class imbalance through the implementation of a paraphrasing method known as synonym replacement.

This works walks through different components. Section 2 presents a review of existing research in the fields of sentiment analysis. Section 3 discusses the dataset used in this study including its source, size and features. Section 4 outlines the proposed model for this research, detailing its architecture and components followed by Section 5 presenting the results of the study including statistics. Finally, Section 6 concludes the summary of the findings and suggesting future research directions.

2. Related Work

This section reviews existing research works on sentiment classification highlighting various methodologies used in the domain of finance. Key themes include embedding techniques, ML model selection, DL model selection and evaluation metrics. The review aims to provide a foundation for current research by synthesizing insights from prior studies and identifying gaps in knowledge.

The goal of the research by Oruganti John Prasad et.al [9] is to provide a comprehensive assessment of different NLP

techniques for online news article sentiment categorization, examining both rule-based and machine-learning methods. It describes how well various natural language processing (NLP) models—such as VADER, TextBlob, SVM, Naive Bayes, RNNs, and transformer-based models like BERT—identify feelings in texts. The study emphasizes the rule-based scoring system of VADER and pre-trained choices suitable for sentiment analysis tasks, such as BERT, RoBERTa, and GPT-2. The article also discusses potential future paths, recommending that financial mood classification to be a key area of study. The paper, "Context-Aware Customer Needs Identification by Linguistic Pattern Mining Based on Online Product Reviews,"[10] introduces a novel method for understanding customer needs from online product reviews. Through linguistic pattern mining, the study extracts and analyzes data from customer reviews to identify specific needs based on the context of product use. This method provides a sophisticated framework that combines semantic embedding and clustering techniques, offering new insights into customer behavior and preferences.The limitations include the inability to compare the proposed context-based approach to prior studies and the lack of testing its applicability to other user-generated content.

Whereas the paper "Negative Review or Complaint? Exploring Interpretability in Financial Complaints"[11] delves into creating a financial complaint detection (CD) system capable of discerning between complaints and non-complaints in financial services. It presents an explainable artificial intelligence (XAI) model employing a dyadic attention mechanism to highlight various text aspects at sentence and word levels. The results indicate XAI's promise in enhancing transparency and confidence in automated financial service systems through interpretable and precise complaint evaluations. An analytics-driven framework integrating deep learning (DL) and multi-criteria decision-making (MCDM) [12] techniques for automated complaint labeling and prioritization. Addressing the challenge of manually categorizing and prioritizing customer complaints, it showcases the effectiveness of transformer-based DL models and MCDM in streamlining the complaint handling process. Experimental results utilizing the BETO model, a BERT variant trained on a substantial Spanish corpus, demonstrate enhanced complaint classification accuracy, leading to improved customer satisfaction and decreased churn rates. Work by Guan D. [13] explored machine learning techniques, specifically Support Vector Machines (SVM) and Naive Bayes classifiers for sentiment analysis of financial online reviews. It emphasizes the significance of understanding sentiment in financial reviews for informed decision-making, presenting a method to classify sentiments into positive, negative, and neutral categories. While both classifiers effectively categorize positive and negative sentiments, Naive Bayes outperforms SVM. On the other hand, the study[14] delves into sentiment classification of customer complaints within Indian banks, utilizing machine learning methodologies. It integrates psycho-linguistic attributes through the LIWC model alongside conventional text analytics techniques such as DTM and Word2Vec. The results indicate that LIWC with classifiers such as Random Forest and Naïve Bayes augments accuracy in sentiment classification, substantiated through statistical validation employing t-tests.

The research work carried out by Lakshmanarao A et.al [15] has implemented various deep learning architectures, including simple RNNs, stacked LSTMs, bidirectional LSTMs, and GRUs to classify sentiments of airline-related tweets. Notable findings include achieving a high accuracy of 93% in distinguishing between sentiments using stacked LSTM models.In addition, the study "Enhancing Sentiment Classification in Twitter Data Through Context-Driven Text Processing and Tweet Embeddings"[16] explains novel text processing techniques and the utilization of skip-gram-based word embeddings to enhance sentiment classification accuracy on Twitter. The authors propose a context-driven method that utilizes word embeddings to capture the distinctive characteristics of tweets. Their findings showcase the effectiveness of context-driven word embeddings in identifying crucial words for tweet classification, surpassing the performance of pre-existing word dictionaries. Aysenur Topbas et.al[17] utilized deep learning models to analyze sentiments expressed in COVID-19-related tweets. Their study compares the performance of RNN and BERT models, with RNN achieving 86.4% accuracy and BERT achieving 83.14% accuracy. The research emphasizes the effectiveness of these models in sentiment analysis on social media, stressing the importance of fine-tuning for optimal results.Another study [18] examines sentiment analysis using a dataset of 15,000 English tweets from ChatGPT users. It employs models like K-Nearest Neighbor (KNN) and Naive Bayes, with Naive Bayes demonstrating superior performance over KNN. A multitask learning framework proposed by Singh A et.al[19] to identify complaints, classify their severity, and recognize underlying emotions and sentiments in tweets. The study details the data collection and annotation process, describes the multitask model structure and compares its performance against various baselines.In contrast,[20] focuses on constructing user profiles using Twitter data for personalized recommendations, particularly in the context of mental health awareness. It involves the classification of health-related tweets and sentiment extraction, utilizing classifiers such as Random Forest, KNN, and Naïve Bayes to analyze sentiments within the tweets.

The gaps in the literature includes the limited exploration of focus on the embedding techniques, limited approach to

addressing the class imbalance issue, lack of further model comparisons and a small test sample dataset. To fill these gaps, this research will compare and use diverse techniques of embedding and the use of hybrid methods and the use of ensemble techniques. Sophisticated techniques including synonym replacement will be applied to enhance the topic's number balance. In this case, several comparative analyses of various categories of the ML and DL models will be prepared to ensure that the current and potential performance benchmarks are well illustrated.

3. Data Description

The FINCORP dataset is consists of grievances concerning financial institutions and their clientele, comprising 6281 complaints sourced from Twitter [21]. Data spans from credit cards to banking issue, providing a comprehensive overview of consumer discontent in the financial sector. It consists of four classes namely; Complaint Label, Severity level, Sentiment and Emotion.

Table 1 lists the attributes and classes along with their labels that need to be categorized and Table 2 contain few sample complaints/suggestions from FINCORP dataset. The total number of words in complaints is 126745 which is an average count of 20.1791. The average number of word in each of the target classes is between 19 to 21.

Table 1. Features in FINCORP

Features	Description	Average word count
Complaints/Opinion	Comprises of complaints or suggestions that the customers have provided	20.17
Target class	**Class Labels**	
Complaint Label	Complaint or non-complaint.(1 or 0)	20.17
Severity level	Accusation, Blame, Disapproval, No explicit reproach, Non-Complaint.	21.15
Sentiment	Negative, Neutral, Positive	19.66
Emotion	Anger,Disgust,Fear, Happiness,Sadness, Surprise, Other	20.70

Table 2. FINCORP Sample Tweet Data by Labels.

Domain	Complaint/Opinion	Complaint Label	Severity level	Sentiment	Emotion
Transaction fail	Can someone tell me if transaction fail between merchant and Bank due to biller issue who responsibilty to contact biller? A bank, A merchant or customer? @ICICIBank_Care @forum_consumer @billdesk #bank #transaction #money #icicibank	1	No explicit reproach	negative	sadness
Netbanking	@ICICIBank_Care @ICICIBank_Care @ICICIBank This is what is happening. When imobile didn't work I reinstalled, since then even activation is not happening. Icicbank website landing page works but netbanking doesn't. With so many people complaining how can you say that there is no problem? https://t.co/vvXNrN5TvI	1	Blame	negative	sadness
Credit Card	1. Got out of credit card debt. 2. Lost 30 pounds (totaling 50 pounds lost since 2019). 3. Made marshmallows (and whole host of other great food) by hand. https://t.co/MO7yluYWMV	0	Non-complaint	positive	other
Debit card	@ishashank08 Hi Shashank! You can deposit funds using Mobikwik via UPI or debit card payments while the wallet top-up is enabled by them. The payment gateway supports limited banks and they are trying to onboard more banks. We shall try to enable UPI asap. Thanks!	0	Non-complaint	neutral	other

Together, the tabular data deepens insights into the dataset's structure, informing the development of precise classification models for Finance Complaint text. Figure 1 gives an overview of class distribution among various classes as mentioned in Table 2.

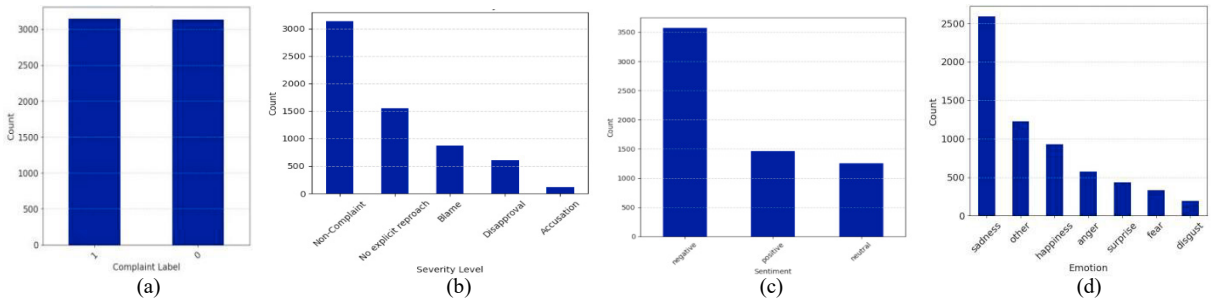


Fig. 1. Distribution of samples Across targets (a) Complaint label (b) Severity level (c) Sentiment and (d) Emotion

4. Methodology

The workflow for classification of sentiment, emotion, severity and complaint label is represented in Fig-3. There are six major steps involved in the proposed methodology; data acquisition, data pre-processing, splitting the data, model building, model evaluation and model explainability which will be discussed in the following section.

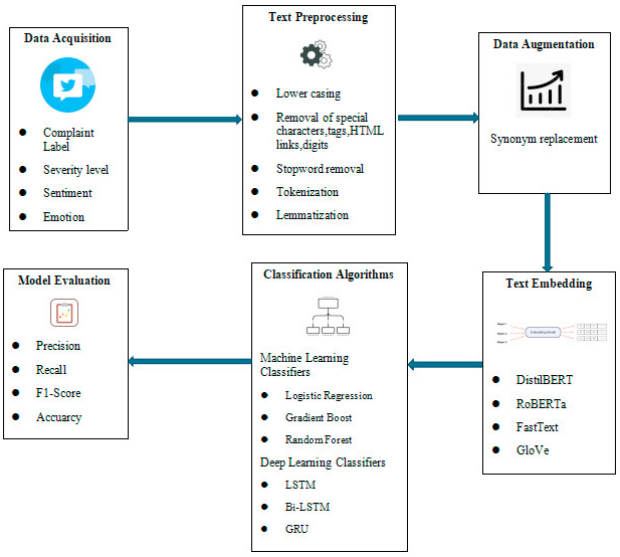


Fig. 2. Proposed Framework for Financial complaint classification

4.1. Text preprocessing

This is a crucial step in any text-based analysis. The first step is converting all the text to lower case which is done to ensure that the same word in different cases is not considered as two different words. The next step is the removal of special characters and digits as they do not contribute to the meaning of the text and can be safely discarded. Furthermore, the data that is being analyzed contains complaint tweets related to the financial domain. These tweets often include links and mentions which are also been removed as they do not contribute to the overall meaning of the text. Stop words, like "the," "a," and "is," are common words that add little to the meaning of a sentence and are eliminated during this phase.

4.2. Data Augmentation

In order to address the issue of an imbalanced dataset, a data augmentation strategy using synonym replacement is implemented. This approach involves substituting original words with their synonyms to generate new, diverse samples. By doing so, the aim is to mitigate the potential negative impact of data imbalance on model performance ensuring that our model can effectively learn from and generalize to various data instances. This technique not only enriches our dataset but also helps the model to better understand and interpret the context which ultimately leads to improved performance and accuracy.

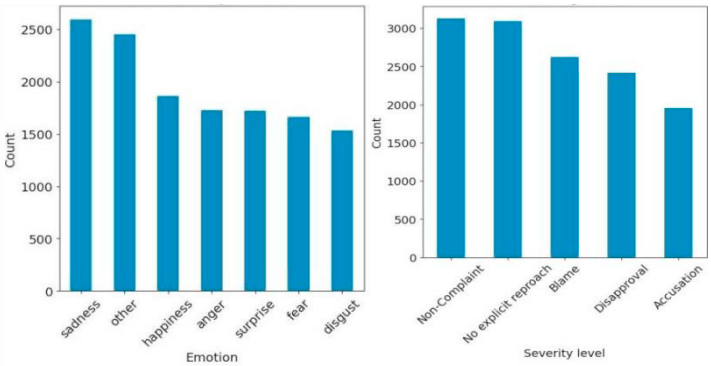


Fig. 3. Emotion and Severity class after data augmentation

It is clear from Fig 1 (b) and (d) that Severity level and Emotion suffers from imbalance in data, augmentation technique has uplifted the distribution through synonym replacement in Fig 3.

4.3. Text Embedding

The conversion of text into numerical form is known as embedding and is a necessary step because computers can only understand numbers. For the analysis of the text data, various embedding models have been incorporated into the corpus. The models examined include Word2Vec [22], Glove, and FastText[23] followed by BERT-based embedding models like DistilBERT and RoBERTa. Word2Vec is a widely-used embedding technique that employs a neural network to learn vector representations of words, offering computational efficient and the ability to process large volumes of text data. In contrast, Glove is an unsupervised learning algorithm that creates word vectors by studying the co-occurrence statistics of words within a corpus. FastText is an extension of Word2Vec that represents each word as an n-gram of characters. This allows it to handle out-of-vocabulary words and provide more accurate representations. BERT [24] based embedding models like DistilBERT [25] and RoBERTa are more recent developments in the field of natural language processing. These models use a transformer based architecture to generate contextualized embeddings of words. This means that the vector representation of a word is dependent on the context in which it appears. These models have been shown to produce state-of-the-art results on a variety of NLP tasks. In this work, the size of embeddings for GloVe and FastText is 100 while the BERT-based embeddings are having a dimension of 768.

4.4. Machine Learning Models

Various machine learning models have been utilized for the classification of four distinct categories, namely Complaint label, Severity level, Emotion, and Sentiment. The models and hyperparameters are tabulated in Table 3 which include logistic regression, SVC, and decision tree. Additionally, ensemble models such as Random forest, gradient boost, AdaBoost, and XGboost have also been used. Each model's performance was assessed and compared to determine the most effective approach for the classification task.

Table 3. Summary of Machine Learning Model Hyperparameters.

Model	Hyperparameter and Values
Logistic Regression	Max_iteration=1000, penalty=L2
Random forest	N_estimators=100, criterion=gini, random_state=42
Gradient Boost	N_estimators=100, learning_rate=0.1, random_state=42, loss=log_loss
Train test split	Test_size= 0.2, stratify=True, random_state=42, shuffle=False

Table 4. Summary of Deep Learning Model Hyperparameters.

Hyperparameter	Value/Shape
Units	128,64
Output Classes	len(label_encoder.classes_)
Loss	'sparse_categorical_crossentropy' / binary crossentropy
Optimizer	'adam'
Metrics	['accuracy']
Epochs	20
Batch Size	32
Dropout	0.2
Activation function	Sigmoid/softmax

4.5. Deep Learning Models

For the classification task, a range of advanced deep learning models are utilized which includes LSTM (Long Short-Term Memory)[26], Bi-LSTM (Bi-directional Long Short-Term Memory), and GRU (Gated Recurrent Unit). LSTM is a type of recurrent neural network (RNN) designed to learn long-term dependencies in data, which makes it ideal for processing sequential information. Bi-LSTM, on the other hand, is an extension of LSTM that can process data in both forward and backward directions, capturing more contextual information. GRU is another variant of RNN that uses gating mechanisms to regulate the flow of information, offering a simpler architecture compared to LSTM while still being effective in handling sequential data. By utilizing these models with the mentioned parameters in Table 4, this work aims to achieve accurate and robust classification results.

4.6. Model Performance Evaluation

In evaluating the performance of the machine learning and deep learning models, a comprehensive assessment was conducted using classification reports. Classification reports were generated for each target class(Compalint label, Severity level, Sentiment and Emotion) to evaluate the performance. These reports included metrics like weighted-average precision, recall, and F1-score, providing a comprehensive view of the models' performance across different domains. A reliable and robust in classifying complaints of various sentiment, severity and emotion snippets across different domains and word embeddings. A spider plot is plotted which is a useful visualization tool for comparing multiple variables across different dimensions and it makes an easier way to identify strengths, weaknesses and patterns. It is also ideal for presenting multidimensional data in a clear and concise manner.

5. Results and Analysis

The performance of various machine learning and deep learning models when combined with different embedding techniques is concluded in this section. To comprehensively assess the effectiveness of these models, precision, recall, accuracy, and F1 score are calculated. These metrics provide insights into the models' ability to correctly classify instances, identify false positives and negatives, and balance precision and recall.

Table 5. Best performance of ML and DL models for classifying Complaint, Severity, Sentiment and Emotion across Embeddings.

Embedding	ML classifiers / DL models	Complaint Label		Severity level		Sentiment		Emotion	
		Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1
DistilBERT	Logistic Regression	0.95	**0.95**	0.71	0.71	0.63	0.63	0.56	0.35
	Gradient Boost	0.92	0.92	0.68	0.67	0.70	0.62	0.56	0.31

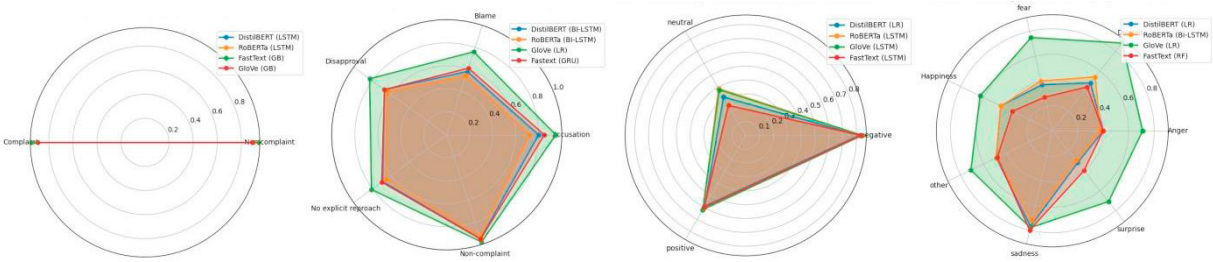
	Random Forest	0.92	0.92	0.66	0.66	0.70	0.58	0.55	0.26
	LSTM	0.95	0.95	0.72	0.71	0.71	0.61	0.52	0.49
	Bi-LSTM	0.95	0.95	0.74	0.74	0.70	0.62	0.52	0.49
	GRU	0.95	0.95	0.72	0.71	0.71	0.62	0.51	0.49
RoBERTa	Logistic Regression	0.94	0.94	0.66	0.66	0.71	0.70	0.45	0.44
	Gradient Boost	0.92	0.92	0.68	0.68	0.71	0.69	0.46	0.45
	Random Forest	0.92	0.92	0.68	0.67	0.70	0.66	0.48	0.46
	LSTM	0.94	0.94	0.69	0.69	0.71	0.70	0.48	0.48
	Bi-LSTM	0.94	0.94	0.70	0.70	0.71	**0.71**	0.49	0.48
	GRU	0.92	0.92	0.68	0.69	0.70	0.70	0.47	0.47
FastText	Logistic Regression	0.93	0.93	0.59	0.58	0.59	0.57	0.39	0.37
	Gradient Boost	0.95	0.95	0.63	0.63	0.63	0.62	0.42	0.41
	Random Forest	0.94	0.94	0.84	0.82	0.61	0.61	0.43	0.42
	LSTM	0.65	0.64	0.73	0.73	0.69	0.59	0.47	0.44
	Bi-LSTM	0.60	0.60	0.73	0.72	0.69	0.58	0.47	0.46
	GRU	0.65	0.64	0.76	0.76	0.68	0.61	0.47	0.47
GloVe	Logistic Regression	0.87	0.87	0.86	**0.86**	0.63	0.63	0.74	**0.74**
	Gradient Boost	0.89	0.89	0.79	0.78	0.62	0.60	0.43	0.41
	Random Forest	0.88	0.87	0.78	0.74	0.62	0.61	0.42	0.40
	LSTM	0.60	0.59	0.68	0.67	0.72	0.59	0.43	0.39
	Bi-LSTM	0.61	0.60	0.69	0.69	0.70	0.61	0.42	0.42
	GRU	0.60	0.59	0.72	0.72	0.69	0.58	0.43	0.39
Baseline[11]	RoBERTa	0.9358	0.92.58	0.7097	0.6788	-	-	-	-

Table 6. Performance comparison with recent literature

Work	**Embedding**	**Model**	**Complaint Label**		**Severity level**	
			Accuracy	F1	Accuracy	F1
Recent Literature[11], 2024	RoBERTa	HAtt + CentralNetAll	0.9358	0.9258	0.7097	0.6788
Proposed work	DistilBERT	Logistic Regression	**0.95**	**0.95**	-	-
	Glove		-	-	**0.86**	**0.86**

Table 5 showcases the performance of the top three ML and DL models combined with the four best-performing embedding techniques across various target classes. DistilBERT embedding is particularly competitive with LSTM achieving an accuracy of 0.71 for sentiment classification. Additionally, LR reaches an accuracy of 0.86 for severity classification using GloVe embeddings. It further highlights the effectiveness of DistilBERT, which attains a remarkable accuracy of 95% in classifying tweets as complaints or non complaints and a 74% accuracy for emotion classification when using GloVe embedding. These results shows a better performance of DistilBERT and the significant impact of embedding techniques on model accuracy for different classification tasks. The combination of advanced embeddings and robust ML/DL models demonstrates the potential for achieving high accuracy in both sentiment and severity classifications in various contexts. Table 6 presents the comparison between baseline work and proposed method in classifying complaint label and severity level. The recent study has achieved F1-Score of 0.92 in classifying a complaint label whereas this study has excelled with a value of 0.95. Additionally, while the

previous work recorded a score of 0.67 for severity classification, this study demonstrated a significant improvement with an F1-score of 0.86.



(a) (b) (c) (d)

Fig. 4. Spider Plots for Best performing Embeddings for (a) Complaint Label; (b) Severity level; (c) Sentiment and; (d) Emotion

The spider plot in Figure 4 represents the performance of four embeddings corresponding to their best two models in classifying the complaints into severity, sentiment and emotion classes respectively. Figure 5(a) and (c) illustrates that the DistilBERT hold the best performing embedding whereas Figure (b) and (d) clearly tells that GloVe has outperformed other embedding techniques . The best models are chosen in terms of F1-score and accuracy across all target classes, highlighting the benefits of combining multiple models. The DL models included LSTM, GRU and Bi-LSTM and ML models include Logistic regression, Gradient Boost and Random forest classifiers. The performance of these models are shown in detail and tabulated in Table 6.

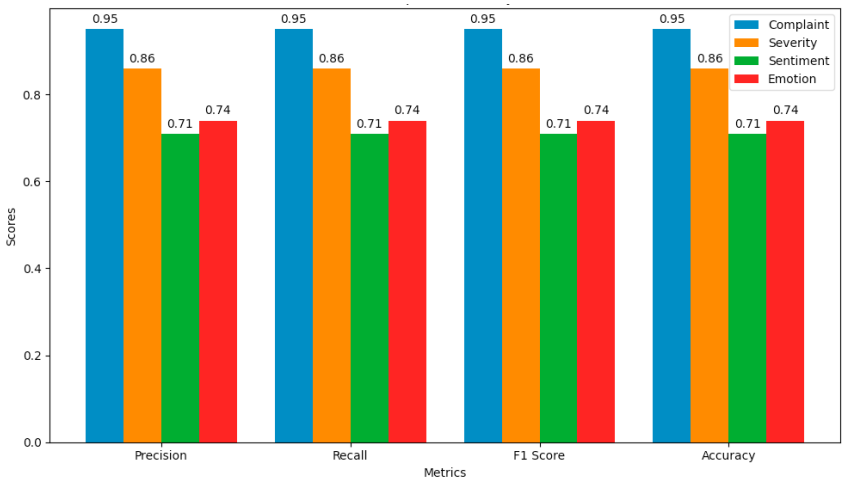


Fig. 5. Classification metrics for Complaint Label; Severity level; Sentiment and Emotion

The Figure 5 illustrates the classification performance metrics for four different classes. Each of the bars corresponds to the Precision, Recall, F1 Score and Accuracy of the models with was achieved in classifying each classes by the proposed model. The Complaint label consistently scored the highest with 0.95 in all metrics, indicating robust performance. The Severity label scored a 0. 86 in all the indices which is relatively good, though lower than that of Complaint label. The Sentiment label receives the lowest score among all the other with 0.71 which can be enhanced in future and the Emotion label depicts a better total score of 0.74 across all metrics. The varying performance levels of different models and embedding techniques is highlighted in this bar plot for the multi-task problem.

	precision	recall	f1-score	support
0	0.95	0.94	0.95	634
1	0.94	0.95	0.95	623
accuracy			0.95	1257
macro avg	0.95	0.95	0.95	1257
weighted avg	0.95	0.95	0.95	1257

(a)

	precision	recall	f1-score	support
Accusation	0.94	0.96	0.95	392
Blame	0.73	0.78	0.76	512
Disapproval	0.85	0.82	0.83	492
No explicit reproach	0.83	0.79	0.81	645
Non-Complaint	0.98	0.98	0.98	604
accuracy			0.86	2645
macro avg	0.86	0.87	0.86	2645
weighted avg	0.86	0.86	0.86	2645

(b)

	precision	recall	f1-score	support
negative	0.84	0.85	0.85	726
neutral	0.44	0.46	0.45	255
positive	0.61	0.57	0.59	276
accuracy			0.71	1257
macro avg	0.63	0.63	0.63	1257
weighted avg	0.71	0.71	0.71	1257

(c)

	precision	recall	f1-score	support
anger	0.78	0.66	0.71	337
disgust	0.90	0.87	0.88	324
fear	0.80	0.71	0.75	328
happiness	0.63	0.63	0.63	358
other	0.69	0.73	0.71	525
sadness	0.76	0.81	0.78	499
surprise	0.69	0.73	0.71	341
accuracy			0.74	2712
macro avg	0.75	0.74	0.74	2712
weighted avg	0.74	0.74	0.74	2712

(d)

Fig. 6. Classification report of best performing models for the multi-task classification (a) Logistic Regression+DistilBERT for Complaint label (b) Logistic Regression+GloVe for Severity level (c) Bi-LSTM+RoBERTa for Sentiment and (d) Logistic Regression+GloVe for Emotion

Classification reports of best performing combination of models and embedding techniques for corresponding target classes is shown in Figure 6. It gives an in depth overview about the performance metrics which includes the weighted and macro average of all the metrics for each class. The performance of each label in their respective class can also be evaluated from this report. It highlights the models strength and hence low score indicates a scope for improvement.

Table 7. Classification results of sample Complaints/opinions (Logistic regression classifier using DistilBERT for Complaint and using GloVe for Severity and Emotion. Bi-LSTM classifier with RoBERTa for Sentiment)

Complaints/Opinions	**Predicted Labels**				**Actual Labels**			
	Complaint Label	Severity level	Sentiment	Emotion	Complaint Label	Severity level	Sentiment	Emotion
Can someone tell me if transaction fail between merchant and Bank due to biller issue who responsibilty to contact biller?A bank,A merchant or customer?@ICICIBank_Care @forum_consumer @billdesk #bank #transaction #money #icicibank	1	No explicit reproach	negative	sadness	1	No explicit reproach	negative	sadness
@LICIndiaForever @Paytmcare @Paytm As part of my LIC premium renewal my balance deducted but payment status shows fail. I have used LIC website where my payment gateways is Paytm. Transaction ID-Â Â 31454836 Kindly address.	1	No explicit reproach	neutral	sadness	1	No explicit reproach	negative	sadness
@ICICIBank_Care This is something very serious nd urgent, i m not able to use imobile app and not able to do any transaction. Urgent solution is required	1	No explicit reproach	neutral	sadness	1	No explicit reproach	negative	anger
@RBI @ICICIBank_Care @ICICIBank @CRED_support @TheOfficialSBI @SBICard_Connect One payment was made with my consent, the another payment seems a fraud transaction both goes to Cred Club.@CRED_club @kunalb11 @ICICIBank @ICICIBank_Care	1	Blame	neutral	sadness	1	Blame	negative	anger

Fraud transaction through CRED for SBI C-Card payment from ICICI Saving account(net banking). Wondering how double payment made for the same cc bill without OTP verification. Amount: 106576.00@RBI@ICICIBank_Care @ICICIBank@CRED_support @TheOfficialSBI@SBICard_Connect	1	No explicit reproach	positive	sadness	1	Accusation	negative	surprise

The classification output for a few tweet samples are given in Table 7. The results suggest that the integration of contextual word embeddings, particularly DistilBERT, has significantly enhanced the performance of both ML and DL models which has achieved an average accuracy of 95%, 86%, 71% and 74% respectively. The minute fine-tuning of hyperparameters and the use of advanced models like Bi-LSTM has helped achieve this result. This also highlights the importance of selecting appropriate embeddings that align closely with the linguistic characteristics and challenges of the target language.

6. Conclusion and Future Scope

This research focuses on sentiment-aware complaint identification in the financial domain using the FINCORP dataset. By leveraging NLP techniques, machine learning, deep learning on various embedding, the study aims to develop models for classifiying sentiment,emotion and severity level in financial complaints. To address the imbalanced data, a data augmentation strategy known as synonym replacement is implemented. Four embeddings that are used for this multi-task are Glove, FastText, DistilBERT and RoBERTa. In sentiment classification, LSTM achieved the highest accuracy of 0.71 using DistilBERT embeddings, followed by Random Forest with 0.70. Furthermore, LR attains a high accuracy of 0.86 for severity classification using GloVe embeddings. DistilBERT achieves an impressive score of 95% accuracy in classifying tweets as complaints or non-complaints and a score of 74% accuracy for emotion classification with GloVe embeddings. These findings highlight DistilBERT's robust performance and the significant influence of embedding techniques on the accuracy of various classification tasks. It can be concluded that no single model excels across all tasks in a multi-task problem, as each model performs differently across various classes. The proposed work signifies a comparative study of various embeddings with a combination of classifiers in classifying the labels. In conclusion, the findings emphasize the value of advanced embedding techniques and robust models in improving sentiment and severity classification performance across different contexts.

Implementing various data augmentation methods help to handle dataset imbalance in a better way. It was observed that there is a significant improvement in accurately classifying the class labels once the data is balanced, which is why this process in explained in this paper. While this work has performed well in its tasks, it also has several limitations. As discussed earlier, each task performs well with a specific combination of embedding and classifier leading to limited generalization and a dependency on chosen embedding. Additionally, the performance scores are specific to this dataset and may not hold when applied to different data. Results could also vary based on the data augmentation techniques chosen. It is suggested that you can incorporate the application of various other large language models (LLMs) as a future scope to further advance the capabilities of the complaint identification system.

References

[1] Osterrieder J. A primer on natural language processing for finance. Available at SSRN 4317320. 2023.
[2] Singh K, Hasan M, Rajendran RP. Opportunities and Challenges of AI/ML in Finance. The Impact of AI Innovation on Financial Sectors in the Era of Industry 5.0. 2023:238-60.
[3] Rangarajan PK, Gurusamy BM, Muthurasu G, Mohan R, Pallavi G, Vijayakumar S, Altalbe A. The social media sentiment analysis framework: deep learning for sentiment analysis on social media. International Journal of Electrical and Computer Engineering (IJECE). 2024 Jun;14(3):1-x.
[4] Venugopalan M, Gupta D, Bhatia V. A supervised approach to aspect term extraction using minimal robust features for sentiment analysis. InProgress in Advanced Computing and Intelligent Engineering: Proceedings of ICACIE 2019, Volume 2 2021 (pp. 237-251). Springer Singapore.
[5] Venkatesh B, Vishwas HN. Real time sarcasm detection on twitter using ensemble methods. In2021 Third International Conference on Inventive Research in Computing Applications (ICIRCA) 2021 Sep 2 (pp. 1292-1297). IEEE.

[6] Mithillesh KP, Supriya M. Multi class sentiment analysis of 5g tweets. In2022 3rd international conference for emerging technology (INCET) 2022 May 27 (pp. 1-5). IEEE.
[7] Abubakar AM, Gupta D, Palaniswamy S. Explainable emotion recognition from tweets using deep learning and word embedding models. In2022 IEEE 19th India Council International Conference (INDICON) 2022 Nov 24 (pp. 1-6). IEEE.
[8] Harjule P, Gurjar A, Seth H, Thakur P. Text classification on Twitter data. In2020 3rd International Conference on Emerging Technologies in Computer Engineering: Machine Learning and Internet of Things (ICETCE) 2020 Feb 7 (pp. 160-164). IEEE.
[9] Prasad OJ, Nandi S, Dogra V, Diwakar DS. A systematic review of NLP methods for Sentiment classification of Online News Articles. In2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT) 2023 Jul 6 (pp. 1-9). IEEE.
[10] Lee J, Jeong B, Yoon J, Song CH. Context-aware customer needs Identification by linguistic pattern mining based on online product reviews. IEEE Access. 2023 Jul 14.
[11] Das S, Singh A, Saha S, Maurya A. Negative Review or Complaint? Exploring Interpretability in Financial Complaints. IEEE Transactions on Computational Social Systems. 2024 Jan 2.
[12] Vairetti C, Aránguiz I, Maldonado S, Karmy JP, Leal A. Analytics-driven complaint prioritisation via deep learning and multicriteria decision-making. European Journal of Operational Research. 2024 Feb 1;312(3):1108-18.
[13] Guan D. Sentiment Classification of Financial Online Reviews Based on Machine Learning Algorithm. In2022 International Conference on Education, Network and Information Technology (ICENIT) 2022 Sep 2 (pp. 289-293). IEEE.
[14] Krishna GJ, Ravi V, Reddy BV, Zaheeruddin M, Jaiswal H, Teja PS, Gavval R. Sentiment classification of indian banks' customer complaints. InTENCON 2019-2019 IEEE Region 10 Conference (TENCON) 2019 Oct 17 (pp. 429-434). IEEE.
[15] Lakshmanarao A, Srisaila A, Kiran TS. Twitter Sentiment Classification with Deep Learning LSTM for Airline Tweets. In2022 8th International Conference on Advanced Computing and Communication Systems (ICACCS) 2022 Mar 25 (Vol. 1, pp. 520-524). IEEE.
[16] Gerogiannis VC, Kanavos A, Antonopoulos N, Bhola A, Acharya B. Enhancing Sentiment Classification in Twitter Data Through Context-Driven Text Processing and Tweet Embeddings. In2023 IEEE 11th Region 10 Humanitarian Technology Conference (R10-HTC) 2023 Oct 16 (pp. 644-648). IEEE.
[17] Topbaş A, Jamil A, Hameed AA, Ali SM, Bazai S, Shah SA. Sentiment analysis for covid-19 tweets using recurrent neural network (rnn) and bidirectional encoder representations (bert) models. In2021 International Conference on Computing, Electronic and Electrical Engineering (ICE Cube) 2021 Oct 26 (pp. 1-6). IEEE.
[18] Ismail WS. Sentiment Analysis of ChatGPT Tweets Using Machine Learning Techniques. In2023 9th International Conference on Optimization and Applications (ICOA) 2023 Oct 5 (pp. 1-5). IEEE.
[19] Singh A, Bhatia R, Saha S. Complaint and severity identification from online financial content. IEEE Transactions on Computational Social Systems. 2023 Feb 17;11(1):660-70.
[20] Joshi M, Bhuvana J. Personalized Recommendation after Classification of Tweets to Predict Depression using Sentiment Analysis. In2023 International Conference on Advances in Computation, Communication and Information Technology (ICAICCIT) 2023 Nov 23 (pp. 827-832). IEEE..
[21] https://github.com/RohanBh23/FINCORP/blob/main/Database/FINCORP.csv
[22] Sugathadasa K, Ayesha B, de Silva N, Perera AS, Jayawardana V, Lakmal D, Perera M. Synergistic union of word2vec and lexicon for domain specific semantic similarity. In2017 IEEE international conference on industrial and information systems (ICIIS) 2017 Dec 15 (pp. 1-6). IEEE.
[23] Khasanah IN. Sentiment classification using fasttext embedding and deep learning model. Procedia Computer Science. 2021 Jan 1;189:343-50.
[24] Zhao A, Yu Y. Knowledge-enabled BERT for aspect-based sentiment analysis. Knowledge-Based Systems. 2021 Sep 5;227:107220.
[25] Akpatsa SK, Lei H, Li X, Obeng VH, Martey EM, Addo PC, Fiawoo DD. Online News Sentiment Classification Using DistilBERT. Journal of Quantum Computing. 2022 Jan 1;4(1).
[26] Kamyab M, Liu G, Adjeisah M. Attention-based CNN and Bi-LSTM model based on TF-IDF and glove word embedding for sentiment analysis. Applied Sciences. 2021 Nov 27;11(23):11255.